

Full-Period Concentration in Mersenne Rule 90

Route-A structural certificate C155

Abstract

For Rule 90 on Mersenne rings $L = 2^r - 1$, the periodic image contains 2^{L-1} states and every period divides L . We prove that exact period L has probability tending to one, the total cycle count is asymptotic to $2^{L-1}/L$, and the average cycle period is asymptotic to L . The neighboring power-of-two family remains nilpotent.

1 Periodic image and proper fixed spaces

On $R_L = \mathbb{F}_2[x, x^{-1}]/(x^L - 1)$, Rule 90 is multiplication by $a = x + x^{-1}$. Frobenius gives $a^{L+1} = a$. Also $\dim \ker a = 1$, so the periodic set is $V = \text{im } a$ with $|V| = 2^{L-1}$, and $g = a|_V$ satisfies $g^L = I$.

For $1 \leq j < L$, put $d = \gcd(j, L)$. The identity $\gcd(U^j - 1, U^L - 1) = U^d - 1$ and Bézout give, after substitution $U = g$ and use of $g^L = I$,

$$\ker(g^j - I) = \ker(g^d - I).$$

A positive-time fixed state already lies in V . Clearing the Laurent multiplier for time d gives $(x^2 + 1)^d + x^d$, of degree $2d$, so

$$\dim \ker(g^j - I) \leq 2d \leq \frac{2L}{3}. \quad (1)$$

The last inequality uses that every proper divisor of odd L is at most $L/3$.

2 Concentration and Burnside

Every state of period less than L belongs to one of the proper-time fixed spaces. Hence

$$\Pr_V(\text{per} < L) \leq \frac{(L-1)2^{2L/3}}{2^{L-1}} \leq 2L2^{-L/3} \rightarrow 0. \quad (2)$$

If C_L is the total number of primitive cycles, Burnside gives

$$C_L = \frac{1}{L} \sum_{j=0}^{L-1} \# \text{Fix}_V(g^j), \quad \left| \frac{LC_L}{2^{L-1}} - 1 \right| \leq 2L2^{-L/3}. \quad (3)$$

The primitive cycles partition V , so their mean length when every cycle is weighted once is $|V|/C_L = 2^{L-1}/C_L$. Dividing by L gives the reciprocal of $LC_L/2^{L-1}$, hence tends to one.

Exact polynomial-gcd and Möbius ledgers through $r = 8$ are finite sentinels. For $L = 2^s$, $a^{L/2} = 0$, so only zero is periodic. The strict tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL). We claim no thermodynamic determinant, target divisor, target functional equation or counting law, arithmetic/local factor, root number, automorphy, natural operator lift, Hilbert–Pólya operator, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.